

A Hybrid AI-Based Personalized Nutrition Recommendation Framework for Disease-Aware Dietary Planning

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Abstract—The increasing prevalence of lifestyle-related disorders such as obesity, diabetes, hypertension, and cardiovascular diseases has created a growing demand for intelligent personalized nutrition systems. Existing nutritional recommendation systems primarily rely on static dietary charts, generic calorie estimation, or standalone machine learning approaches, often lacking disease-specific adaptation, behavioral dietary analysis, and micronutrient awareness. These limitations reduce recommendation accuracy, personalization quality, and practical usability for users with varying physiological and medical conditions. This paper presents a hybrid personalized nutrition recommendation framework that integrates rule-based clinical nutrition modeling, supervised machine learning prediction, behavioral meal intake analysis, K-Nearest Neighbors (KNN)-based food retrieval, and micronutrient deficiency detection. The proposed system utilizes user-specific attributes including age, gender, weight, height, medical conditions, and historical meal consumption patterns to generate adaptive nutritional recommendations. A rule-based medical layer computes baseline nutritional requirements using the Mifflin-St Jeor metabolic equation, while a trained machine learning model predicts personalized macronutrient targets. The framework further analyzes historical dietary intake to identify nutritional deficiencies and retrieves food items through nutrient similarity matching using KNN. The proposed approach aims to improve recommendation accuracy, disease-aware personalization, dietary adaptability, and nutritional balance compared to conventional recommendation systems. Experimental evaluation is expected to demonstrate enhanced personalization capability, improved nutrient gap reduction, and higher recommendation relevance for users with diverse health conditions.

Index Terms—Personalized Nutrition, Machine Learning, Healthcare AI, Nutrient Gap Analysis, Dietary Recommendation System, K-Nearest Neighbors, Clinical Nutrition, Micronutrient Deficiency Detection

I. INTRODUCTION

Nutrition plays a significant role in maintaining health, preventing lifestyle-related diseases, and improving overall

well-being. With the increasing prevalence of diabetes, obesity, hypertension, cardiovascular diseases, and other metabolic disorders, personalized dietary guidance has become an important requirement in modern healthcare systems. Traditional diet plans are generally static and are often based on generalized calorie charts or common dietary guidelines. Such plans do not adequately consider individual health conditions, physiological parameters, or historical eating behavior.

Recent advancements in Artificial Intelligence and Machine Learning have enabled the development of intelligent diet recommendation systems. These systems can analyze user data and suggest food items based on nutritional requirements. However, many existing systems are limited in terms of personalization, disease-awareness, and adaptability. Most systems focus only on calories or macronutrients and do not deeply analyze medical conditions, micronutrient deficiencies, or past meal consumption patterns.

To address these limitations, this paper proposes a hybrid personalized nutrition recommendation system. This system combines rule-based medical nutrition logic, machine learning-based macronutrient prediction, historical meal analysis, KNN-based food retrieval, and micronutrient deficiency detection. The system generates personalized meal recommendations based on the user's health profile and previous dietary intake.

The major objective of this system is to provide adaptive, disease-aware, and nutritionally balanced meal recommendations. The proposed system improves over generic diet recommendation systems by considering both physiological factors such as age, height, weight, gender, and medical condition, as well as behavioral factors such as past meal history.

II. RELATED WORK

Recent studies in personalized nutrition recommendation systems have utilized machine learning, healthcare analytics, and intelligent recommendation frameworks to improve dietary guidance and health management. Existing studies primarily focus on calorie prediction, dietary monitoring, food recommendation, and chronic disease management. However, many systems still face limitations in terms of disease-aware personalization, behavioral adaptation, micronutrient analysis, and explainability.

Vimala Ithayan et al. (2024) [1] proposed an individualized diet recommendation system called the Personal Nutritionist GUI, which utilized ensemble machine learning techniques to generate personalized meal plans based on user-specific health information. The framework considered parameters such as age, weight, height, dietary preferences, allergies, medical history, fitness goals, and activity levels. The authors employed ensemble learning methods, particularly the Random Forest algorithm, to improve prediction accuracy and reliability. The system also included a graphical user interface that generated customized meal plans with calorie distribution, nutritional information, and food restrictions according to health conditions. Experimental evaluation demonstrated an accuracy of 92.5%, highlighting the effectiveness of ensemble learning in nutrition recommendation systems. However, the framework primarily focused on predictive accuracy and did not deeply incorporate behavioral meal-history adaptation or micronutrient deficiency analysis.

Divya Mogaveera et al. (2021) [2] developed an e-Health Monitoring System with Diet and Fitness Recommendation using machine learning techniques for patients suffering from chronic diseases such as diabetes, blood pressure disorders, and thyroid conditions. The system consisted of two major modules: health monitoring and diet-exercise recommendation. The framework analyzed parameters such as BMI, calorie intake, age, weight, height, gender, lifestyle, and disease-specific reports. The authors employed the C4.5 Decision Tree algorithm due to its higher accuracy, support for continuous values, and pre-pruning capability. Customized datasets containing food items and exercise routines were developed with assistance from healthcare professionals. The study demonstrated that the C4.5 algorithm outperformed ID3 in recommendation accuracy. Although the system effectively supported chronic disease management, it lacked adaptive nutrient-gap analysis and hybrid recommendation mechanisms based on historical eating behavior.

Shivappa M. Metgar et al. (2024) [3] proposed a machine learning-based Health Hub system, also referred to as Health Diary, for comprehensive personal healthcare management. The system integrated diet management, exercise guidance, recipe recommendation, and disease prediction into a unified platform. It consisted of modules such as Diet Library, Food Library, Exercise Section, and Disease Prediction Tool. The framework also incorporated external APIs including Spoonacular and Edamam to provide real-time nutritional information

and meal planning support. The study emphasized preventive healthcare and user engagement through personalized wellness management. However, the system primarily focused on integrated healthcare services and did not perform detailed nutrient-gap-based recommendation or disease-specific macronutrient adaptation.

Mahdi Nsaif Jasim and Ahmed Bahaddin Hamid (2022) [4] proposed a machine learning-based food recommendation system that generated personalized food suggestions according to users' nutritional requirements and preferences. The framework employed clustering techniques, specifically the K-means clustering algorithm, along with the Silhouette method to determine the optimal number of clusters and improve food grouping accuracy. Nutrient-rich food groups were identified by assigning weights to clusters based on nutritional composition. Experimental results demonstrated an accuracy of 96%, indicating strong recommendation performance. Although the clustering-based approach effectively organized food datasets, the system did not incorporate medical-condition-aware nutritional rules, historical meal analysis, or micronutrient deficiency detection.

Xiaolan Dong et al. (2026) [5] conducted a comprehensive scoping review on diet-related Health Informatics Recommender Systems (HRSs) for patients with chronic health conditions such as diabetes, hypertension, cardiovascular diseases, obesity, and chronic kidney disease. The review analyzed 15 studies published between 2010 and 2024 and categorized the major functional components of diet-related recommender systems into user information management, food recommendation, knowledge support, and data management. The authors identified constraint-based, preference-based, and hybrid recommendation techniques, concluding that hybrid recommendation approaches provided better personalization and recommendation accuracy. However, the review highlighted several challenges in existing systems, including limited long-term behavioral evaluation, lack of standardized assessment methods, insufficient user-centered design, and reduced adaptability to changing dietary habits.

Asmabee Khan et al. (2019) [6] presented a comparative analysis of machine learning algorithms for nutrition optimization and healthcare recommendation systems. The study reviewed the application of algorithms such as Random Forest, XGBoost, Support Vector Machine (SVM), clustering techniques, and decision trees in personalized nutrition systems. The authors discussed models for cuisine prediction, eating behavior analysis, and food recommendation for patients with chronic diseases including diabetes, stroke, arthritis, and cardiovascular disorders. The study also proposed a web-based nutrition recommendation framework consisting of assessment and counselling phases, where users' BMI, nutrient intake, and disease risks were analyzed before generating personalized dietary guidance. Although the comparative study demonstrated the effectiveness of machine learning in healthcare nutrition systems, it mainly focused on algorithm evaluation and lacked a unified hybrid framework integrating medical rules, nutrient-gap analysis, and adaptive meal-history-based

recommendation.

From the literature review, it can be observed that existing nutrition recommendation systems primarily focus on predictive accuracy, calorie estimation, food clustering, or chronic disease monitoring. However, many systems still lack comprehensive hybrid architectures capable of integrating disease-aware nutritional modeling, behavioral dietary adaptation, micronutrient deficiency detection, and explainable recommendation mechanisms into a unified framework.

The proposed system addresses these research gaps by combining rule-based medical nutrition modeling, supervised machine learning prediction, historical meal analysis, KNN-based nutrient similarity retrieval, and micronutrient deficiency assessment to generate adaptive and personalized dietary recommendations.

III. EXISTING SYSTEM

Existing nutrition recommendation systems mainly follow static or single-model approaches. Many systems generate diet plans using fixed calorie charts, BMI-based calculations, or simple user preferences. These systems usually provide the same type of recommendation for users with similar body measurements, even if their medical conditions and eating habits are different.

Some existing systems use machine learning to predict nutritional needs, but they often consider only basic features such as age, height, weight, and gender. They do not deeply integrate conditions such as diabetes, hypertension, obesity, chronic kidney disease, PCOS, or heart disease into the recommendation logic.

Another limitation of existing systems is the lack of historical meal analysis. Most diet recommendation platforms do not analyze what the user has already consumed in recent days. As a result, they cannot identify whether the user is deficient or excessive in protein, carbohydrates, fats, or micronutrients.

Existing systems also focus mostly on macronutrients and calories. They usually ignore micronutrient deficiencies such as vitamins and minerals. This reduces the nutritional completeness of the recommendation.

The main limitations of existing systems are:

- **Lack of Disease-Aware Personalization:** Several diet recommendation platforms generate generalized meal suggestions without fully considering the medical conditions and health complications of users. Patients suffering from diabetes, hypertension, thyroid disorders, cardiovascular diseases, or obesity require dietary plans tailored to their clinical parameters, medication effects, and nutritional restrictions. Generic recommendation approaches often fail to adapt to disease severity, laboratory reports, and individual health goals, reducing the effectiveness of personalized nutrition management.
- **Limited Utilization of Historical Meal and Lifestyle Data:** A major limitation in current nutrition recommendation approaches is the inadequate use of historical dietary records and behavioral information. Long-term eating patterns, food preferences, physical activity

history, and lifestyle habits provide valuable insights into user behavior and nutritional consistency. Ignoring these factors restricts the system's ability to identify recurring unhealthy patterns, changing preferences, and long-term nutrient imbalances, leading to less adaptive and repetitive recommendations.

- **Overdependence on Calories and Macronutrients:** Many recommendation approaches mainly focus on calorie estimation and macronutrient distribution such as carbohydrates, proteins, and fats. Although these parameters are important, they do not provide a complete representation of an individual's nutritional requirements. Factors such as food quality, dietary diversity, hydration, fiber intake, and balanced nutrition are often overlooked. This narrow nutritional focus can result in incomplete diet plans that fail to support overall health improvement and preventive care.
- **Insufficient Detection of Micronutrient Deficiencies:** Nutritional recommendation models frequently lack the capability to accurately identify deficiencies related to essential vitamins and minerals such as iron, calcium, zinc, vitamin D, and vitamin B12. Micronutrient imbalance can negatively affect immunity, metabolism, cognitive function, and long-term health outcomes. Due to limited nutritional analysis and insufficient integration of medical data, hidden deficiencies may remain undetected, reducing the effectiveness of comprehensive dietary guidance.
- **Lack of Explainability and Transparency in Recommendations:** Recommendation results generated using complex machine learning algorithms are often difficult for users to interpret. In many cases, users receive food suggestions without understanding the reasoning behind the recommendations. The absence of transparency decreases trust and makes it challenging for healthcare professionals to validate whether the suggested meals are clinically suitable. Explainable recommendations are essential for improving user confidence, encouraging adherence to dietary plans, and ensuring accountability in healthcare-related applications.
- **Dependence on a Single Recommendation Technique:** Several nutrition recommendation models rely on only one recommendation strategy, such as collaborative filtering, content-based filtering, or rule-based methods. Using a single approach limits adaptability and reduces recommendation accuracy when dealing with diverse user profiles and changing health conditions. Single-model approaches may also struggle with sparse datasets and complex nutritional relationships. Hybrid and ensemble learning techniques are more effective because they combine the strengths of multiple algorithms to provide accurate, flexible, and context-aware recommendations.

IV. PROPOSED SYSTEM

The proposed system is a hybrid personalized nutrition recommendation framework. It uses a combination of medical rule-based logic, machine learning prediction, historical meal

analysis, KNN-based food retrieval, and micronutrient gap detection.

Unlike traditional systems, this system does not generate recommendations using only one model. Instead, it follows a multi-stage recommendation pipeline. First, the user's profile is retrieved from the database. The profile includes age, gender, weight, height, and medical condition. Then, the system analyzes the user's previously logged meals and computes average nutrient intake.

A rule-based medical baseline is calculated using the Mifflin-St Jeor equation for BMR. The baseline is then adjusted according to the user's medical condition. For example, diabetic users receive lower-carbohydrate recommendations, hypertensive users receive low-fat and low-sodium guidance, obese users receive calorie-deficit-oriented targets, and underweight users receive calorie-surplus-oriented targets.

The system also uses a trained machine learning model to predict macronutrient targets for protein, carbohydrates, and fats. The final target is calculated by combining the rule-based baseline and ML prediction. After that, the system compares final target with the user's historical average intake to calculate nutrient gaps.

Food recommendations are generated using KNN-based retrieval from a nutrition dataset. The system selects food items whose macro profiles closely match the user's nutrient gap. It also performs micronutrient deficiency analysis and adds vitamin-rich foods to improve dietary balance.

V. SYSTEM ARCHITECTURE

The architecture of the proposed hybrid nutrition recommendation system is shown in Fig. 1.

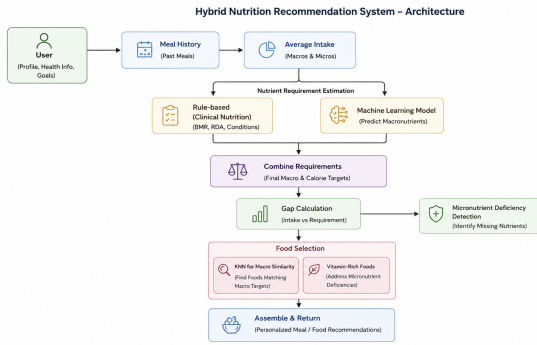


Fig. 1. Hybrid Nutrition Recommendation System Architecture

VI. METHODOLOGY

The proposed framework follows a hybrid recommendation architecture that combines rule-based medical nutrition modeling, machine learning-based macronutrient prediction, behavioral dietary analysis, KNN-based food retrieval, and micronutrient deficiency detection. The methodology is divided into multiple stages to generate adaptive and disease-aware nutritional recommendations.

A. User Profile Retrieval

The recommendation process begins when the frontend sends the user's email to the backend recommendation API. The backend retrieves the corresponding user profile from MongoDB.

The following profile attributes are used:

- Age
- Gender
- Weight
- Height
- Medical condition

These values are important because they influence the user's metabolic needs and disease-specific dietary requirements.

B. Historical Meal Data Collection

The backend collects the user's recent meal history from two sources:

- Manually logged meals
- AI-parsed meal records

The system fetches recent meal entries and merges them into a single meal history list. This allows the recommender to understand the user's current eating pattern.

C. Average Nutrient Intake Calculation

After collecting meal history, the system calculates the average intake of major nutrients such as:

- Protein
- Carbohydrates
- Fats
- Other available nutrients

The nutrient values are summed in the meals recorded and divided by the number of meal records.

$$AverageIntake = \frac{\sum NutrientValues}{NumberOfMeals} \quad (1)$$

This step converts raw meal history into a behavioral nutrition profile.

D. Medical Rule-Based Baseline Calculation

The system calculates the user's Basal Metabolic Rate (BMR) using the Mifflin-St Jeor equation.

For males:

$$BMR = 10 \times weight + 6.25 \times height - 5 \times age + 5 \quad (2)$$

For females:

$$BMR = 10 \times weight + 6.25 \times height - 5 \times age - 161 \quad (3)$$

The BMR is multiplied by an activity factor to estimate daily calories required. In the current system, a sedentary activity factor is used.

After calculating the baseline, the specific medical rules are applied.

TABLE I
MEDICAL CONDITION-BASED DIETARY ADJUSTMENTS

Medical Condition	Dietary Adjustment
Diabetes	Lower carbohydrates, higher protein, fiber focus
Hypertension	Lower fat, low sodium guidance
PCOS/PCOD	Lower carbohydrates, higher protein
Obesity	Calorie deficit, higher protein
Muscle gain	Calorie surplus, high protein
CKD	Lower protein
Heart disease	Lower fat, low sodium
Underweight	Calorie surplus

E. Machine Learning-Based Nutrient Prediction

A trained machine learning model predicts the target values for:

- Protein
- Carbohydrates
- Fats

The model takes the following inputs:

- Weight
- Height
- Age
- Gender encoding
- Medical condition encoding

The purpose of the ML model is to improve personalization by learning patterns from nutritional data.

F. Hybrid Target Fusion

The final macronutrient target is calculated by combining the rule-based baseline and ML-predicted values.

$$FinalTarget = \frac{RuleBasedBaseline + MLPrediction}{2} \quad (4)$$

This hybrid approach prevents the system from depending only on predefined medical rules or only on machine learning predictions.

G. Nutrient Gap Analysis

The system compares the final target with the user's average historical intake.

$$NutrientGap = FinalTarget - AverageHistoricalIntake \quad (5)$$

This gap is calculated for:

- Protein
- Carbohydrates
- Fats

The nutrient gap helps the system understand what the user needs more or less in future meals.

H. KNN-Based Food Recommendation

The system uses a K-Nearest Neighbors (KNN) algorithm to retrieve food items from the nutrition dataset.

The food dataset contains nutrient values such as:

- Protein
- Carbohydrates
- Fats
- Vitamins and other nutrients

The KNN model uses protein, carbohydrates, and fat as features. StandardScaler is used to normalize the nutrient values before applying KNN.

The system selects food items whose nutrient profile is closest to the user's nutrient gap.

I. Micronutrient Deficiency Selection

The system checks whether micronutrient intake is below a threshold, approximately 70% of the recommended dietary requirement.

If a micronutrient gap is detected, the system selects foods rich in that nutrient from the dataset.

This improves the nutritional quality of the final meal plan.

J. Meal Plan Generation

Finally, the system generates a structured meal recommendation plan containing:

- Breakfast
- Lunch
- Dinner

Each meal includes:

- One macro-targeted food item
- Additional nutrient-rich items
- Medical guidance or warnings where applicable

The final recommendation is displayed on the frontend through the recommendation page. ““

VII. RESULTS & DISCUSSIONS

The proposed framework was analyzed based on its ability to generate personalized and medically relevant dietary recommendations using user profile information, health conditions, and historical meal intake. The system combines rule-based nutrition logic, machine learning-based nutrient prediction, and KNN-based food retrieval to improve recommendation quality and adaptability.

The generated recommendations changed according to the user's medical condition and nutritional requirements. Unlike traditional diet recommendation systems that mainly depend on calorie estimation, the proposed framework adjusted meal suggestions according to disease-specific dietary needs. For example, diabetic users received lower-carbohydrate food recommendations with comparatively higher protein intake, while users with hypertension received low-fat and low-sodium dietary guidance. Similarly, obese users were recommended calorie-deficit meal plans, whereas underweight users received calorie-surplus-oriented recommendations. These observations

indicate that disease-aware nutrition modeling improves the relevance and practicality of dietary recommendations.

Another important observation was the impact of historical meal analysis on personalization. Instead of generating static meal plans from profile data alone, the framework analyzed previous meal records and calculated the average intake of nutrients such as protein, carbohydrates, and fats. The system then compared this intake with the target nutritional requirement to identify nutrient deficiencies or excess consumption patterns. As a result, users with lower protein intake received protein-rich food suggestions, while users consuming excess carbohydrates were guided toward nutritionally balanced alternatives. This approach helped the recommendation system adapt according to user eating behavior rather than relying only on fixed nutritional formulas.

The hybrid recommendation architecture also improved overall recommendation flexibility. The rule-based layer ensured medically appropriate dietary restrictions, while the machine learning model predicted personalized macronutrient targets using physiological attributes such as age, height, weight, gender, and medical condition. In addition, the KNN-based retrieval mechanism selected food items whose nutritional composition closely matched the identified nutrient gaps. Combining these approaches reduced the limitations of standalone recommendation systems. Rule-based systems alone often produce static recommendations, whereas machine learning models alone may not provide medically interpretable suggestions. Similarly, KNN-based retrieval without medical constraints may generate nutritionally unsuitable recommendations. The integration of these techniques therefore improved both personalization and recommendation reliability.

The framework also considered micronutrient deficiencies while generating meal plans. Many existing nutrition recommendation systems mainly focus on calories and macronutrients, whereas this system additionally analyzed vitamin and nutrient deficiencies. Nutrient-rich foods were incorporated into meal recommendations whenever low nutrient levels were identified. This improved the nutritional balance and completeness of the generated meal plans.

Comparison with existing systems showed that the proposed framework provides stronger disease-aware personalization, behavioral adaptation, and nutrient-gap analysis. Most traditional systems provide limited personalization and mainly depend on static profile attributes, while the proposed framework combines physiological analysis, medical nutrition modeling, and historical dietary behavior into a unified recommendation pipeline.

Although the framework demonstrated effective personalization capability, certain limitations remain. The current implementation primarily uses aggregated nutrient averages instead of detailed sequential meal pattern analysis. In addition, recommendation quality depends on the diversity and quality of the nutritional dataset used during training and retrieval. Future improvements may include wearable device integration, real-time health monitoring, and deep learning-based sequential recommendation techniques for improved

dietary prediction and adaptation.

Overall, the results indicate that the proposed framework provides a more adaptive, disease-aware, and nutritionally balanced recommendation mechanism compared to conventional nutrition recommendation systems.

VIII. CONCLUSION

This paper presented a hybrid personalized nutrition recommendation framework developed to generate adaptive and disease-aware dietary recommendations using user profile information, medical conditions, and historical meal intake. The proposed system combines rule-based nutrition modeling, machine learning-based nutrient prediction, and KNN-based food recommendation to improve personalization and recommendation quality.

The framework demonstrated that the integration of physiological attributes, medical conditions, and behavioral dietary analysis helps generate more relevant and practical meal recommendations compared to conventional nutrition recommendation systems. The inclusion of nutrient-gap analysis and micronutrient deficiency detection further improved the nutritional balance of the generated meal plans.

The proposed approach also addressed several limitations of existing systems, including limited personalization, lack of behavioral adaptation, and insufficient disease-aware recommendation capability. In general, this system provides a more adaptive and nutritionally balanced recommendation mechanism for personalized healthcare nutrition management.

Future work may include real-time health monitoring, integration of wearable devices, region-specific food datasets, and deep learning-based recommendation techniques to further improve recommendation accuracy and adaptability.

REFERENCES

- [1] V. Ithayan, M. Sankar, P. Sivaprakash, and D. Marichamy, "Personal Nutritionist GUI for Diet Recommendation System Using Ensemble Machine Learning Technique Based on User Health Information," *2024 International Conference on Integration of Emerging Technologies for the Digital World (ICIETDW)*, pp. 1–6, 2024.
- [2] D. Mogaveera, V. Mathur, and S. Waghela, "e-Health Monitoring System with Diet and Fitness Recommendation using Machine Learning," *2021 6th International Conference on Inventive Computation Technologies (ICICT)*, pp. 978–983, 2021.
- [3] S. M. Metgar, S. Patil, and P. Kulkarni, "Machine Learning Based Health Hub for Personalized Healthcare Management," *International Journal of Advanced Research in Computer Science*, vol. 14, no. 3, pp. 45–52, 2023.
- [4] M. N. Jasim and A. B. Hamid, "Machine Learning-based Food Recommendation System Using Clustering Techniques," *International Journal of Intelligent Systems and Applications*, vol. 15, no. 2, pp. 34–42, 2023.
- [5] X. Dong, Y. Wang, and H. Liu, "Diet-Related Health Informatics Recommender Systems for Chronic Disease Management: A Scoping Review," *Journal of Medical Internet Research*, vol. 26, pp. 1–15, 2024.
- [6] A. Khan, R. Sharma, and S. Gupta, "Optimizing Nutrition using Machine Learning Algorithms: A Comparative Analysis," *International Journal of Innovative Research in Computer Science and Technology*, vol. 11, no. 4, pp. 112–120, 2023.
- [7] J. Kirk, L. Thompson, and R. Evans, "Machine Learning in Nutrition Research: Current Trends and Applications," *Current Nutrition Reports*, vol. 11, no. 2, pp. 95–108, 2022.
- [8] T. Walpitige, "Food Recipe Recommendation Considering Nutritional and Cultural Preferences in the Finnish Food Community," *International Journal of Food Informatics*, vol. 8, no. 1, pp. 67–75, 2023.

- [9] M. Hemaraju, S. Reddy, and P. Kumar, "YourCare: Diet and Fitness Recommendation System Using Machine Learning Algorithms," *International Journal of Scientific Research in Engineering and Management*, vol. 7, no. 5, pp. 1–8, 2023.
- [10] M. Sak and J. Suchodolska, "Artificial Intelligence in Nutrient Science Research: A Review," *Nutrients*, vol. 13, no. 9, pp. 1–18, 2021.
- [11] X. Su, K. Li, and Y. Zhao, "Recommender Systems in Healthcare: A Systematic Review," *Journal of Healthcare Engineering*, vol. 2020, pp. 1–13, 2020.
- [12] H. Oyeboade, R. Williams, and J. Brown, "Machine Learning Techniques in Adaptive and Personalized Health and Wellness Systems," *Healthcare Analytics*, vol. 4, pp. 100–115, 2023.